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Course: BIOS331: Microbial Genomics

Identifying Adaptive vs. Biochemical-Consequence Variants in Laboratory *E. coli*

Introduction

Background: The parental *E. coli* strain lacks two proteins and grows slowly in rich media. Two independently evolved lines recover faster growth. Our goal was to separate changes that are likely just biochemical effects due to the missing proteins from changes that are likely adaptive.

Data: The reference genome considered was *E. coli* K-12 (NC_000913) which was originally a GenBank file. The reads were compressed (gunzipped) FASTQ for Parent, Evolved line 1, and Evolved line 2

Methods

Reference Preparation: Because BWA requires a FASTA reference, the GenBank file was converted to FASTA before indexing. This was done using the tool *EMBOSS seqret* on Galaxy. Then, this served as an input as we index the reference using *BWA-MEM2 indexer*.

Read Mapping: *BWA-MEM2* on Galaxy was used to map the Parent reads, Evolved strain 1 reads, and Evolved 2 strain reads. The outputs were coordinate-sorted BAMs containing mapped reads for each sample.

Variant Calling: *BBTools* was used to call variants with the reference FASTA and each sample's BAM. The output was per-sample VCFs.

Functional Annotation: The SnpEff annotation tool on Galaxy repeatedly errored, so we ran SnpEff locally. We built a local SnpEff database from the Genbank file. Then, we ran SnpEff eff on each VCF to add gene names, impacts (HIGH/MODERATE/LOW), coding consequences (synonymous, missense, frameshift, etc.), and predicted effects. The output was annotated VCFs (per sample) and simple tabular summaries.

Copy Number Analysis: Conducted using VarScan and Control FREEC

Interpretation

To distinguish between Biochemical and Adaptive variants, we use the following schema:

- **Biochemical-consequence set**
 - Those variants common to Parent and both evolved lines (same locus, same allele) are biochemical. We reason so because these changes likely arose as direct biochemical/physiological consequences of the engineered loss of the two proteins in the parental background and were simply carried forward. They are unlikely to explain the improved growth of the evolved lines, because they are not exclusive to each line.
- **Adaptive set**
 - For Evolved line 1, variants present in Evolved Strain 1 that are absent in the Parent are adaptive in nature. We consider these changes as they arose during the adaptation of Evolved strain 1 and were absent before the adaptation. However, it is important to note that not all private variants are necessarily adaptive. Some may have arisen randomly through neutral drift or background mutational processes, without conferring any growth advantage. To separate these from truly adaptive changes, a literature search is required, focusing on the genes and pathways involved, to determine whether they have known roles in metabolism, stress response, or growth regulation.
 - For Evolved Line 2, the same principle applies: variants absent in the Parent but present in Evolved Strain 2 are considered candidate adaptive changes. As with Strain 1, some may be adaptive, while others may be random passengers with little effect, requiring functional context from the literature.

Note: We also attempted to identify intersections between Evolved Strain 1 and Evolved Strain 2 to see whether parallel adaptations emerged. One such set was found and is discussed below. However, we see that two evolved strains carry distinct private mutations, which could represent different genetic solutions to the same growth limitation. This is consistent with the idea of multiple evolutionary paths leading to recovery of fitness under the same selective pressure.

Findings

Variant Analysis:

Number of Parent Variants: 2

Number of Variants in Evolved Strain 1: 7

Number of Variants in Evolved Strain 2: 7

Set of Variants that are biochemical consequences:

	Locus (genomic coordinates)	Ref	Alt	Gene	Variant type	Location
1	NC_000913.3:2173360–2173362	ACC	A	gatC (b2092)	Intragenic (noncoding) – deletion	Intragenic (gatC); also near gatR/gatD (upstream) and gatB/A/Z/Y, fbaB (downstream)
2	NC_000913.3:3560455	C	CG	glpR (b3423)	Intragenic (noncoding) – insertion	Intragenic (glpR); near glpD (upstream), malT/rtcR/glpG/E/yzgI/glpP (downstream)

Set of Likely Adaptive Variants in Evolved Strain 1:

	Locus (genomic coordinates)	Ref	Alt	Gene	Variant type	Location
1	NC_000913.3:1411925	T	A	ttcA (b1344)	Synonymous (p.Thr8Thr)	CDS (synonymous)
2	NC_000913.3:1411926	G	C	ttcA (b1344)	Missense (p.Thr8Arg)	CDS (missense)
3	NC_000913.3:1411933	G	C	ttcA (b1344)	Missense (p.Gln6Glu)	CDS (missense)
4	NC_000913.3:3471741	C	T	fusA (b3340)	Missense (p.Ala592Thr)	CDS (missense)
5	NC_000913.3:4296380	A	ACG	gltP–yjcO (intergenic)	Intergenic – insertion	Intergenic (between gltP and yjcO); downstream of nrfD/E/F/G and gltP

It is likely the variant in *fusA* is the main adaptive one in Strain 1. *fusA* encodes Elongation Factor-G, the essential GTPase that drives tRNA–mRNA translocation and contributes to ribosome recycling. EF-G kinetics set the elongation rate, a primary control on protein production and hence the growth in rich media. We observe that Strain 1 carries *fusA* A592T. Its allele frequency is high ($>0.98 \approx 0.984$), meaning it is essentially fixed, and is strong evidence of selection. It's possible that this mutation could possibly result in higher elongation throughput per ribosome. A study ([Holley et al. 2022, Antimicrob Agents Chemother.](#)) identified a specific *fusA* missense mutation (A563V – similar variant locus to the modified protein in the strain considered) in *Neisseria gonorrhoeae* that conferred increased gentamicin resistance; it did not impair in vitro growth or protein synthesis in the tested conditions. Although the study doesn't explicitly claim faster growth, the absence of growth impairment suggests that some *fusA* mutations may maintain or potentially enhance growth under selective pressures.

Set of Likely Adaptive Variants in Evolved Strain 2

	Locus (genomic coordinates)	Ref	Alt	Gene	Variant type	Location
1	NC_000913.3:1411925	T	A	ttcA (b1344)	Synonymous (p.Thr8Thr)	CDS (synonymous)

2	NC_000913.3:1411926	G	C	<i>ttcA</i> (b1344)	Missense (p.Thr8Arg)	CDS (missense)
3	NC_000913.3:1411933	G	C	<i>ttcA</i> (b1344)	Missense (p.Gln6Glu)	CDS (missense)
4	NC_000913.3:2867374	G	C	<i>rpoS</i> (b2741)	Missense (p.Leu60Val)	CDS (missense)
5	NC_000913.3:4296380	A	ACG	<i>gltP-yjcO</i> (intergenic)	Intergenic – insertion	Intergenic (between <i>gltP</i> and <i>yjcO</i>); downstream of <i>nrfd</i> /E/F/G and <i>gltP</i>

Note: There was a non-variant call at genomic coordinate 1378565 (A→A, implying REF=ALT) which was omitted in this table.

Studies have shown that when RPOS is inactivated/knocked out, there is an increased TCA cycle activity and altered metabolic flux in *rpoS* mutants, which supports faster growth during the exponential phase relative to wild-type *E. coli* ([Rahman et al. 2006, *Biotechnol Bioeng*](#)). This makes the *rpoS* a potential gene of interest in this study. Also, the allele frequency was about 0.9915 for *rpoS*, indicating most reads have this variant, implying most members of the population do have it as well, highlighting its importance.

On Variants common to Both Evolved strains and absent in Parent Strain

Both evolved lines carry the same tight triad in *ttcA* (positions 1411933, 1411926, 1411925; two missense at aa 6 and 8, plus a synonymous at aa 8). Getting independent changes in *ttcA* twice strongly suggests *ttcA* is under selection in the background. It follows that the two missense (Gln6Glu, Thr8Arg) mutations are the most plausible adaptive changes. However, the allele frequency for these variants across both strains is between 0.55–0.7, meaning only ~50–70% of reads carry the mutation, meaning it may not be essential for survival. This suggests a mixed population, which suggests either:

- It's still sweeping through the culture, or
- It's neutral and coexisting with the wild-type

We couldn't find any literature on *ttcA* mutations helping in the growth of *E. coli* strains in the lab. However, a hypothesis is as follows: *ttcA* is a tRNA thioltransferase that modifies cytidine in tRNA, a process that is non-essential under rich laboratory conditions but metabolically costly. These mutations in *ttcA* likely reduce its activity or expression, saving Fe–S and sulfur resources while modestly relaxing translation fidelity. This shift could lower maintenance burden and improve proteome allocation, and hence yield a growth advantage in the evolved *E. coli* strains.

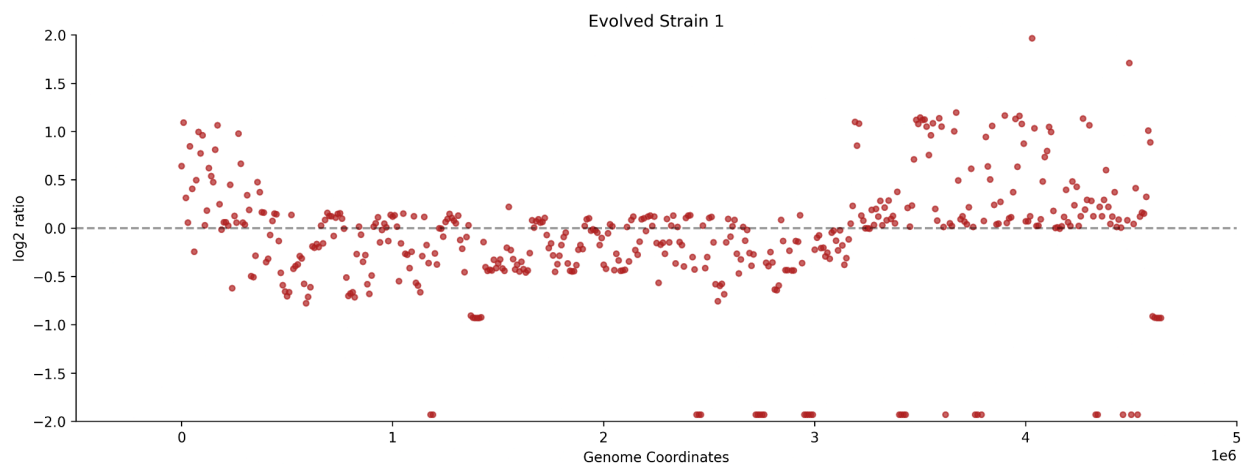
Similarly, for the parallelism shown by the *gltP-yjcO* intergenic insertion, we can explain why it arises in two ways:

- The site sits between the 3' ends of *glpP* and *yjcO* and is annotated only as downstream/intergenic. A short local repeat/microstructure can promote small +2 bp slippage insertions, so the same neutral change can recur independently and then hitchhike to fixation with the true drivers (*fusA* in Strain 1, *rpoS* in Strain 2).
- Somehow, this +2 bp shift confers some selective advantage. However, there doesn't seem to be literature on this.

It is also plausible that the parallel changes observed in both evolved strains represent biochemical by-products of the evolutionary process rather than direct drivers. In this view, such variants are absent in the parent & present in both evolved lines may reflect mutational hotspots or structural predispositions in the genome. They may arise during reproduction (or even adaptation) and then persist. Consequently, because the parent strain reproduced far less and did not undergo adaptation, it had a much smaller mutation supply and fewer opportunities for such variants to arise or reach detectable frequency. The evolved strains, in contrast, underwent many more generations under selection, making recurrent hotspot events more likely to appear and persist. In short, absence in the parent and presence in both evolved lines can reflect exposure to the evolutionary process rather than causality

Copy Number Analysis:

Copy Number Analysis on Evolved Strain 1: This part was conducted via Control FreeC. Plot indicates duplications in the 0~0.5 Mbp range and ~3.2Mbp to 4.5Mbp range.



From the Control FREEC output, we observe 2 distinct duplication segments

- ~0–0.37 Mbp ($\approx$ 1-2 copies on average)
- ~3.19–4.6 Mbp ($\approx$ 1-2 copies)
- ~3.47–3.66 Mbp ($\approx$ 2-3 copies: Increased duplications)

These results indicate substantial genome remodeling with both amplifications as compared to the parent. The gains potentially influence growth and stress-response traits.

Comparing the two evolved lines highlights different evolutionary solutions to the same growth problem.

- **Strain 1** improved fitness partly through large-scale structural changes: Duplications that adjust gene dosage across megabase segments of the chromosome.
- **Strain 2** seems to have achieved faster growth without altering copy number, relying instead on small point mutations

Copy Number Analysis on Evolved Strain 2: Conducted using both VarScan and Control Freec. No significant duplications or deletions were observed. Plot differs in density from the one above due to differences in Segmentation criteria in VarScan and Control Freec.

